DG INTERNS HUB — WEEK 4 TECHNICAL REPORT

COMPLETE SIEM & SECURITY MONITORING COMPILATION

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Deliverable File: DG_Interns_Hub_Week_4_Complete_SIEM_Report_Gunjan.pdf

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1. EXECUTIVE SUMMARY

This technical report documents the comprehensive studies, practical lab foundations, threat detection rules, and security monitoring architectures completed during Week 4 of the DG Interns Hub Cybersecurity Internship. As modern corporate networks face increasing volumes of sophisticated cyber threats, Security Operations Center (SOC) analysts rely on Security Information and Event Management (SIEM) systems to collect, normalize, correlate, and investigate security telemetry across endpoints and network perimeters.

This report consolidates previously completed Splunk home lab authentication investigations, Week 4 theoretical architecture principles (Splunk vs Wazuh, SOAR, EDR, Event Correlation), a catalog of 30 production-ready SPL search queries, comprehensive brute-force detection techniques mapped to MITRE ATT&CK T1110, custom Snort NIDS local rules (SID 1000001–1000005) mapped to T1046, Snort-to-SIEM integration pipelines, lab troubleshooting records (Oracle VirtualBox Kali VM disk expansion and Splunk trial license reactivation), and a structured 10-step SOC incident response workflow.

2. PREVIOUS PRACTICAL EXPERIENCE WITH SPLUNK ENTERPRISE

Prior to Week 4, practical experience was gained by setting up a Splunk Enterprise home lab to monitor and investigate Windows Security Event logs. Key activities included:

- **Installation & Configuration:** Deploying Splunk Enterprise on Windows, navigating the Splunk Web interface, and configuring file inputs.
- **Windows Security Logging:** Ingesting and parsing the `WinEventLog:Security` sourcetype, analyzing successful logons (Event ID 4624) and failed logons (Event ID 4625), and examining logon types (e.g., Type 2 interactive, Type 3 network, Type 10 RDP).
- **Basic SPL Practised:** Utilizing core commands (`search` , `stats` , `eval` , `table` , `fields` , `where` , `sort` , `dedup` , `timechart` , `top` , `count` , `rename` , `rex`).

Important Environment Limitation: Field names and index names vary across deployments depending on ingestion agents (e.g., Winlogbeat vs. Universal Forwarder). Queries use standard field names like `TargetUserName` or `Source_Network_Address` , which require proper field extractions.

3. WEEK 4 — SIEM THEORY AND TECHNICAL CONCEPTS

- **What Is SIEM?:** Combines SIM (historical log archiving and compliance) and SEM (real-time event correlation and alerting) into a centralized SOC command center.
- **SIEM vs. SOAR vs. EDR:** SIEM provides enterprise-wide log aggregation and visibility; EDR focuses on host telemetry, process tree tracking, and process isolation; SOAR automates incident response playbooks and ticket orchestration.
- **Splunk vs. Wazuh:** Splunk is a commercial, enterprise-grade data platform using SPL and indexing buckets; Wazuh is an open-source XDR/SIEM platform built on OpenSearch with native FIM and vulnerability detection.
- **Event Correlation & Alerting:** Connecting disparate events using temporal and relational logic. Threshold alerts track volume spikes, while correlation searches combine multi-source indicators to reduce false positives.

4. SPL SEARCH QUERIES — PRACTICAL REFERENCE (CATALOG PART 1)

Below is a curated reference of essential SPL queries organized across fundamental and authentication categories:

1. SEARCH ALL EVENTS (LAB SCOPE)

```
index=*
```

Objective: Raw indexing verification. **Use Case:** Testing pipeline connectivity. **Limit:** High CPU overhead in production.

2. FILTER SPECIFIC INDEX

```
index=wineventlog
```

Objective: Scope search to Windows logs. **Use Case:** Targeted investigations.

3. WINDOWS SECURITY LOG SOURCETYPE

```
index=wineventlog sourcetype="WinEventLog:Security"
```

Objective: Isolate OS security audit data.

4. FILTER FAILED LOGINS (EVENT ID 4625)

```
index=wineventlog EventCode=4625
```

Objective: Capture failed authentication attempts.

5. TIME-BOUNDED SEARCH

```
index=wineventlog EventCode=4625 earliest=-24h latest=now
```

Objective: Restrict search window to past 24 hours.

6. FORMAT FAILED LOGINS TABLE

```
index=wineventlog EventCode=4625 | table _time, TargetUserName, Source_Network_Address, WorkstationName
```

Objective: Display timestamp, user, IP, and machine in a clean table.

7. COUNT FAILED LOGINS BY USER

```
index=wineventlog EventCode=4625 | stats count by TargetUserName | sort - count
```

Objective: Identify targeted user accounts.

8. COUNT FAILED LOGINS BY SOURCE IP

```
index=wineventlog EventCode=4625 | stats count by Source_Network_Address | sort - count
```

Objective: Identify attacking external or internal hosts.

4. SPL SEARCH QUERIES — PRACTICAL REFERENCE (CATALOG PART 2)

16. BRUTE-FORCE THRESHOLD RULE (>10 FAILURES IN 15 MIN PER COMPUTER)

```
index=wineventlog EventCode=4625 earliest=-15m
| bin _time span=15m
| stats count by _time, Computer, TargetUserName, Source_Network_Address
| where count > 10
| sort - count
```

Objective: Core automated alerting rule for brute-force detection. Uses `bin` for time slicing and `where` for threshold evaluation.

20. HOURLY AUTHENTICATION TREND ANALYSIS

```
index=wineventlog (EventCode=4624 OR EventCode=4625)
| timechart span=1h count by EventCode
```

Objective: Visualize success vs. failure ratios over time.

25. COMPROMISED ACCOUNT DETECTION (FAILURE FOLLOWED BY SUCCESS)

```
index=wineventlog (EventCode=4625 OR EventCode=4624)
| stats count(eval(EventCode=4625)) as Failures, count(eval(EventCode=4624)) as Successes by Ta
rgetUserName, Source_Network_Address
| where Failures > 10 AND Successes > 0
```

Objective: High-severity alert indicating brute-force success.

30. DASHBOARD TOP ATTACKING IPS

```
index=wineventlog EventCode=4625 | top limit=5 Source_Network_Address
```

Objective: Populate SOC monitoring dashboard pie chart with top 5 offending IPs.

5. BRUTE-FORCE DETECTION & MITRE ATT&CK T1110

MITRE T1110 (Brute Force): Adversaries systematically guess credentials across sub-techniques T1110.001 (Password Guessing), T1110.002 (Password Spraying), T1110.003 (Credential Stuffing), and T1110.004 (Account Cracking). Monitoring Windows Event IDs 4625 (failures), 4740 (lockouts), and 4624 (successes) allows SOC analysts to detect automated credential attacks.

6. SNORT IDS — LOCAL RULES CONFIGURATION

Snort is an open-source NIDS/NIPS that inspects network packets in real time. Custom local rules (stored in `local.rules`) enable SOC analysts to write bespoke detection signatures for internal environments.

Five Practical Custom Snort Rules

RULE 1: ICMP PING DETECTION

```
alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"LOCAL-RULE ICMP Ping Request Detected"; itype:8; sid:1000001; rev:1;)
```

RULE 2: INBOUND SSH CONNECTION ATTEMPT

```
alert tcp $EXTERNAL_NET any -> $HOME_NET 22 (msg:"LOCAL-RULE Inbound SSH Connection Attempt"; flow:to_server,established; flags:S; sid:1000002; rev:1;)
```

RULE 3: UNENCRYPTED HTTP WEB REQUEST OUTBOUND

```
alert tcp $HOME_NET any -> $EXTERNAL_NET 80 (msg:"LOCAL-RULE Unencrypted HTTP Web Request Outbound"; flow:to_server; content:"GET"; depth:3; sid:1000003; rev:1;)
```

RULE 4: NMAP SYN SCAN DETECTION (MITRE T1046)

```
alert tcp $EXTERNAL_NET any -> $HOME_NET any (msg:"LOCAL-RULE Possible Nmap TCP SYN Scan Detected"; flags:S; window:1024; sid:1000004; rev:1; classtype:attempted-recon;)
```

MITRE Mapping: Maps to **T1046 (Network Service Discovery)**. Indicates TCP reconnaissance scanning.

RULE 5: SUSPICIOUS CLEARTEXT PASSWORD TRANSMISSION

```
alert tcp $HOME_NET any -> $EXTERNAL_NET any (msg:"LOCAL-RULE Suspicious Cleartext Password Transmission"; content:"password="; nocase; sid:1000005; rev:1; classtype:suspicious-traffic;)
```

7. SNORT AND SIEM INTEGRATION

Snort writes alert telemetry to `/var/log/snort/alert`. A SIEM forwarder or Wazuh agent tails this log file, normalizes fields (`src_ip`, `dst_ip`, `sid`), and correlates network alerts with Windows authentication logs (e.g., matching a Snort port scan alert with subsequent SMB connection attempts).

8. LAB ENVIRONMENT & TROUBLESHOOTING NOTES

Environment Specs: Windows 10 Host, Oracle VirtualBox, Kali Linux VM (4 GB RAM, x86_64, Version 2026.2).
Troubleshooting Observations: The original 25 GB VirtualBox disk was expanded to 40 GB. GParted was used to inspect partitions, and swap was temporarily disabled. Splunk Enterprise on Windows encountered an expired trial license on 10 October 2026; the licensing group was successfully changed to Free, and a restart was performed to resolve licensing restrictions.

9. SOC ANALYST WORKFLOW & 10. LEARNING OUTCOMES

SOC Workflow: 1. Ingestion → 2. Normalization → 3. Monitoring → 4. Detection → 5. Correlation → 6. Triage → 7. MITRE Mapping → 8. Containment → 9. Escalation → 10. Reporting.
Student Reflection: Working through Week 4 cemented my understanding of SIEM architectures, SPL querying, brute-force telemetry, custom Snort rule writing, and incident response workflows.

11. CONCLUSION & REFERENCES

This report demonstrates a solid foundation in SOC monitoring and threat detection.
References: Official Splunk Documentation, Snort 3 User Manual, MITRE ATT&CK Framework.